

After good earnings, does the extra return wait for a breakout?

Lei Kam Fai (Woody). Quant Team Recruitment, Round 1. 14 September 2026. Data: Massive API and SEC EDGAR.

The answer in one paragraph

A popular trading rule says: after a great earnings report, wait until the stock closes above its earnings-day high, then buy. I tested whether the extra return really arrives after that "breakout". I used 14,770 US earnings days from 2005 to 2026 where the stock jumped into the top 10% of reactions. **It does not.** After a breakout, the stock earns 0.9 basis points a day above the market; the same stocks earn 1.8 a day over the same days whether or not they broke out. Waiting for the breakout earned *less* than buying the next morning (about 0.9% less per trade before 2015, about the same after). Buying the next morning earned about 1.4% over 40 days before 2015 and nothing since, and trading costs of about 1% per round trip wipe that out. **The idea is rejected.**

1. The question

Some stocks keep rising for weeks after a strong earnings report. Traders call this drift. The question is not *how much* drift there is, but *when* it arrives: steadily from the next day, or mostly after the stock first closes above the high of the earnings day?

2. The idea, and the trap in it

Two rules on the same stock, same earnings day, same 40-day window:

Rule	Buys	Sells
A	next morning at the open	close of day 40
B	the morning after the first close above the earnings-day high	close of day 40

If drift arrives only after the breakout, B should earn more than A. But look at a made-up stock:

Day	Price	What happens
Earnings day	close 100, high 104	big jump
Day 1 open	100	A buys
Day 3 close	105	first close above 104
Day 4 open	105	B buys
Day 40 close	110	both sell

A makes +10%. B makes +4.8%.

Why Rule A must beat Rule B on breakout cases

On any case where the stock breaks out, both rules hold the same position from B's entry to the end. The only difference is that A also held the stock from day 1 until B's entry. So:

A's return = (1 + run-up) x (1 + B's return) - 1, where the run-up is the price move from A's buy price to B's buy price.

B buys only after a close *above the day-o high*, and A bought at the day-1 open, which is almost always *below* that high. So the run-up is positive in 88% of breakout cases (average +3.8%), and A's return is larger than B's on almost every one of them. In the example: $(1 + 5\%) \times (1 + 4.8\%) - 1 = 10\%$.

B can therefore only win *on average* if the stocks that *never* break out fall a lot, because on those B earns 0 while A takes the loss. In our data, 61% of cases break out (A +4.2%, B +0.3%) and 39% do not (A -5.1%, B 0). "B beats A" is a statement about the losers, not about *when* drift arrives. This trap shaped the whole design: the main test in section 6 compares only the part of the return that a breakout trader actually receives.

3. Data

Source	What	Size	Used for
Massive: Daily Market Summary	open, high, low, close, volume for every US stock, each weekday 2004 to Sep 2026	5,988 calls, 23 files	prices, returns, SPY benchmark
Massive: All Tickers	every ticker, live and dead, with company ID (CIK)	36,588 rows	link filings to prices; keep delisted stocks
Massive: Splits	every split and stock dividend	27,541 rows	adjust prices ourselves
SEC EDGAR submissions (free, public)	every 8-K filing with the exact time the SEC accepted it	321,077 earnings 8-Ks, 9,679 companies	date each earnings event

Prices were downloaded *unadjusted* and split-adjusted by us, so price filters use the real price of the day. Our adjustment matches the vendor's own adjusted prices (tested). Raw data is not in the repository (licence); all summary tables are.

Why EDGAR. Massive's 8-K endpoints give a filing *date* but no time, and their earnings tags start in 2020 and miss most releases. EDGAR gives the acceptance time to the second. It is public and free, and the time it shows is the time the news became public.

Cleaning and checks

Data	Problem	What we did	Where
Prices	vendor's adjusted prices rescale history to today's share count, so a \$5 filter would look at the wrong price	downloaded unadjusted prices; applied split factors ourselves; checked against the vendor's factors (all tickers) and adjusted prices (split tickers)	prices.py, test_prices.py, test_api_crosscheck.py
Prices	duplicate rows across pages	dropped duplicate (ticker, date) rows	panel.py
Prices	holidays and missing days	trading calendar = days SPY traded; a return is missing if the previous day's bar is missing	panel.py
Prices	ETFs, warrants, preferreds, OTC names mixed in	kept common stocks (type CS) on NYSE, Nasdaq, NYSE American, plus SPY	panel.py, events.py
Prices	bad bars	tested: high $\geq$ open and close, low $\leq$ open and close in 99.8% of bars; closes positive; about 252 days per year	test_prices.py
Tickers	one symbol reused by two companies over time	each ticker record is valid only between the previous namesake's delisting and its own	panel.py
Tickers	companies with several share classes	kept the most liquid class per event	events.py
Filings	amendments and repeat filings	dropped 8-K/A; one event per company per day 0; dropped follow-ups within 20 sessions	events.py
Filings	times are in UTC; SEC dates shift after 17:30 New York time	used the acceptance time converted to New York time, not the filing date	edgar.py, test_edgar.py
Filings	8-K can lag the press release	measured: biggest move on day 0 for 64% of events, day +1 for 20%; reported as a limit	timing_validation.csv
Events	reaction needs two consecutive bars	required a bar on day -1 and day 0; 1 event had no day-1 open and was dropped from return tables	events.py, test_events.py
Events	nothing from the future	tested: threshold built only from earlier quarters; filters use day -1 data; every buy is after its signal	test_events.py
Vendor	two Massive endpoints disagree on about 1% of days by more than 1%	used one endpoint (Daily Market Summary) throughout; noted	decision log 01:10

4. How each event was built

1. Take every 8-K with Item 2.02 (earnings release).
2. **Day 0** = the first session that could react: filed before 09:30 New York time, same day; filed after 16:00, next trading day; filed during the session, set aside as "unclear" (used only in a check).
3. Link the company to its stock through the CIK, including stocks that no longer trade.
4. Keep US common stocks on NYSE, Nasdaq or NYSE American, price at least \$5, average daily dollar volume at least \$1 million, at least 120 days of history. All measured *before* day 0.

5. Measure the day-0 reaction: return from the day before to day 0, minus SPY.
6. Keep a "strong report" if the reaction is in the top 10% of the *previous four quarters*. This threshold was knowable on the day; ranking within the same quarter would use the future.

Result: 14,770 strong reports, 3,116 stocks, 86 quarters. Median reaction +12.4%. 22% are stocks that later delisted. Check on the dating: for 64% of events the biggest move falls on our day 0, for about 20% one day later (an 8-K filed hours after the press release). This noise hits Rules A and B equally.

step	events left
edgar_item202_rows	321077
8k_not_amended	318081
unique_cik_day0	316021
with_candidate_ticker	311923
matched_to_prices	288469
after_dedupe_20d	271534
exchange_filter	271460
plus_price_filter	226721
plus_liquidity_filter	184493
plus_history_filter	181927
plus_not_intraday	145809
in_sample_period_filtered	142358
selected_top_decile_trailing	14770
selected_fixed_5pct	26611
selected_zo_ge_2	35517
selected_bottom_decile	14771

5. The rules and the controls

Days are trading days after day 0. H = holding length (40). W = longest wait for a breakout (10).

Rule	Buys at the open of	Condition	What it tests
A	day 1	none	plain drift
B	day k+1	first close above the day-0 high on day k (k up to W); else no trade	the breakout idea
C	day 3	none (day 2 is B's typical signal day)	is it just the waiting?
D1 / D2 / D3	day k+1	first close above the day-0 close / 3% above the high / the day-1 high	is the earnings-day high special?

All sell at the close of day H. Returns are "abnormal": the stock's return minus SPY's over the same span. If a stock stops trading inside the window, it is sold at its last price (17 cases).

6. What we measure

- **Main test.** For each breakout case, the abnormal return *per day* from its entry to day H, compared with what *all* strong-report stocks in the same quarter earned per day over those *same* days. This compares like with like and avoids the trap in section 2.
- **Second test.** Rule B's 40-day return minus Rule A's.
- **Uncertainty.** Earnings come in four seasons a year, so events are not independent. Every number carries a 95% range from resampling whole quarters (86 of them, 5,000 times). If the range includes zero, the effect is not distinguishable from zero.
- **Two settings only.** W (5, 10, 15, 20) and H (20, 40, 60). The default (10, 40) was fixed before any result. 2005 to 2014 picks the best cell; 2015 to 2026 reports it. All other choices (the \$5, the \$1 million, the top 10%) were fixed once and later checked, not tuned.

7. Results (W = 10, H = 40)

What	2005 to 2026	2005 to 2014	2015 to 2026
Rule A, 40-day return	+ 0.55% (-0.50 to +1.59)	+1.43% (+0.57 to +2.35)	+0.12% (-1.29 to +1.55)
Rule B, 40-day return	+0.21%	+0.51%	+0.06%
B minus A	-0.34% (-0.87 to +0.20)	-0.92% (-1.41 to -0.46)	-0.06% (-0.77 to +0.67)
After breakout, per day	0.9 bp		
Same days, all cases, per day	1.8 bp		
Main test: difference per day	-0.9 bp (-1.5 to -0.2)	-1.5 bp (-2.4 to -0.6)	-0.6 bp (-1.4 to +0.3)

bp = basis point = 0.01%. Ranges are 95% confidence intervals.

Three things to see.

1. **Drift is small and old.** Buying the next morning earned +1.4% before 2015 and nothing after (Figure 1). This matches published work saying the drift has faded.
2. **Nothing extra after the breakout.** 61% of cases broke out within 10 days, 45% of them on day 1. After the entry they earned 0.9 bp a day; the normal path was 1.8 bp a day. If anything, slightly less (Figure 2). A gap of 1 bp a day is about 0.4% over the window.
3. **Waiting lost money.** B trailed A in every period, clearly before 2015. Why: breakout cases earned +4.2% under A but only +0.3% under B (the run-up was +3.8%); the 39% that never broke out lost 5.1% under A while B sat out.

Controls. C (buy on day 3, no signal) earned +0.45%. D1, D2, D3 earned +0.40%, +0.08%, +0.12%. Every "wait for a higher close" rule behaves like B. The earnings-day high is not a special level.

cases	n	share of cases	Rule A	Rule B	run-up B skipped
all events	14769	1	0.00546673	0.00207921	0.037913
breakout within W	8974	0.607624	0.041938	0.00342187	0.037913
no breakout within W	5795	0.392376	-0.0510118	0	nan

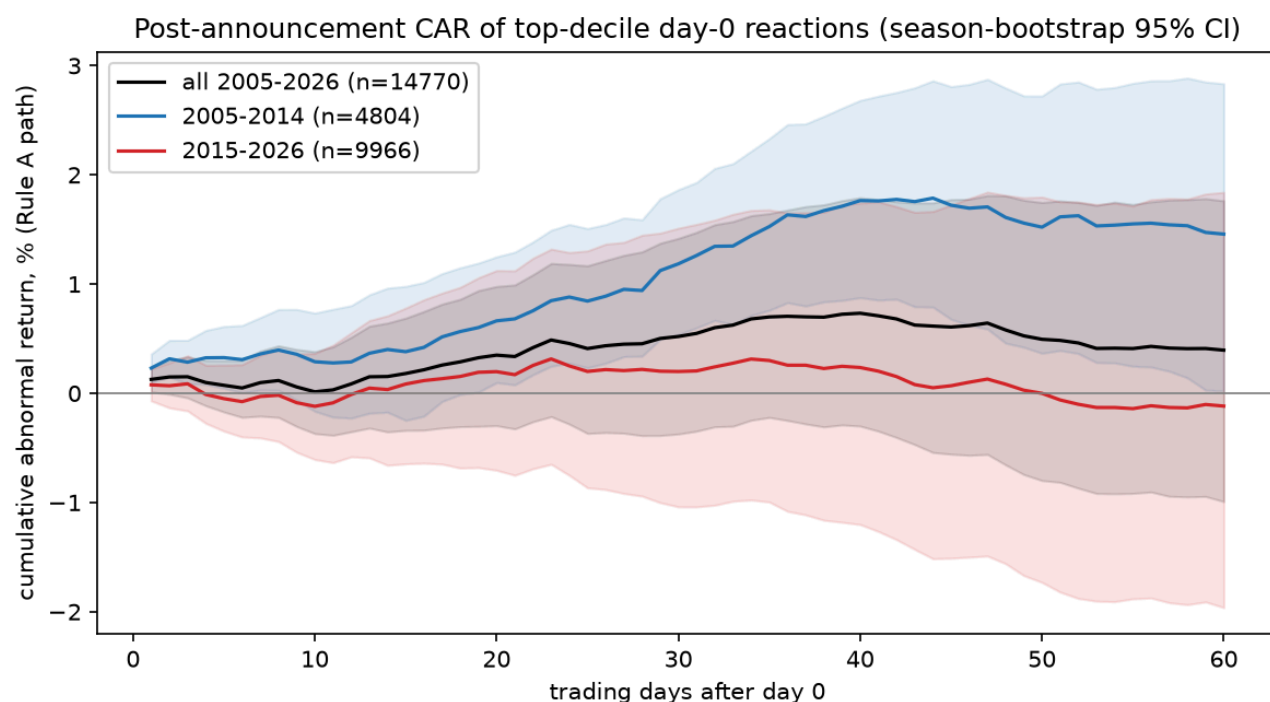


Figure 1. Return above the market after a strong report (Rule A), with 95% ranges.

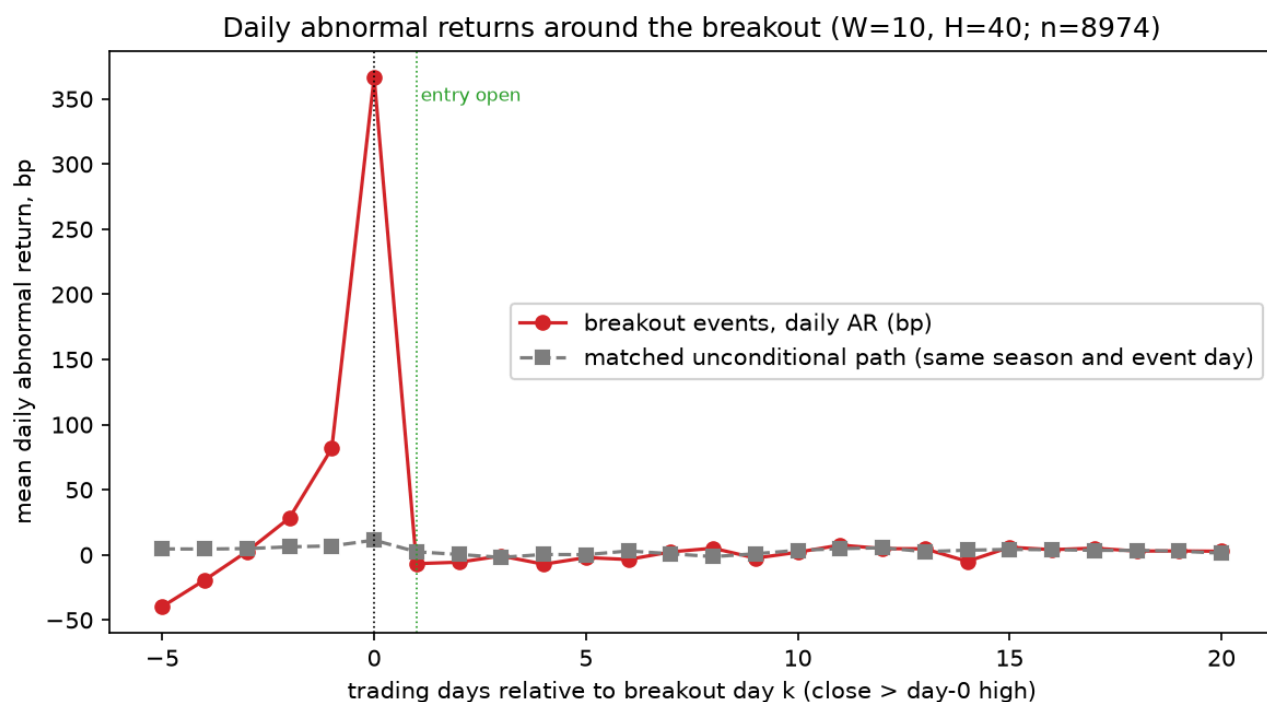


Figure 2. Daily return above the market around the breakout day. Red: breakout cases. Grey: all cases on the same days.

How to read Figure 2. Day 0 here is the *breakout* day, not the earnings day. The red spike at 0 is large by definition: a breakout is a day the stock rose. Everything right of the green line is what a Rule

B trader actually gets. It sits on or just under the grey line.

Same result in calendar time. Averaging all open positions each day, the A book earned 2.9 bp a day ($t = 2.5$), the B book 1.1 bp, and B minus A was -1.8 bp a day ($t = -2.9$). Before 2015: A 5.2 bp ($t = 3.8$); after: 0.8 bp ($t = 0.5$).

8. Does the choice of W and H matter? No.

B minus A is negative in all 12 cells for 2005 to 2014 (the range is below zero in 11) and not distinguishable from zero in all 12 cells for 2015 to 2026. The main test stays between -1.8 and +0.3 bp a day everywhere. The cell picked on 2005 to 2014 was $W = 20$, $H = 60$; on 2015 to 2026 it gives -0.1 bp a day (-0.7 to +0.5) and B minus A of +0.17% (-0.58 to +0.91).

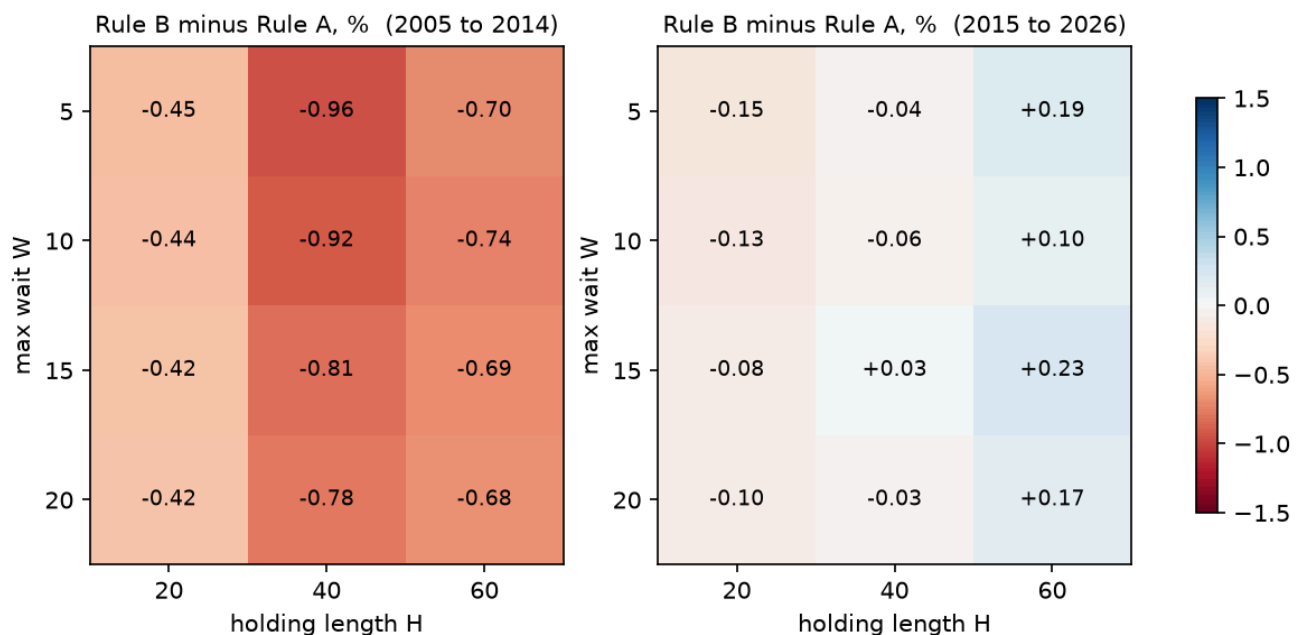


Figure 3. B minus A for every W and H. Left: 2005 to 2014. Right: 2015 to 2026.

9. Checks

Every check ran through the same code and is logged in `logs/experiment_record.csv`.

Check	B minus A	Extra drift after breakout (bp/day)	Meaning
Base	-0.34%	-0.9 (-1.5 to -0.2)	
Ignore day-1 breakouts	-0.37%	-0.9	not a day-1 artefact
Buy at the signal close, not next open	-0.41%	-1.2	faster fills do not help
"Strong" = at least +5% (26,610 cases)	-0.18%	-0.7	same picture
"Strong" = 2 standard deviations (35,515)	-0.07%	-0.5	same picture

Check	B minus A	Extra drift after breakout (bp/day)	Meaning
Top 15% / top 5% instead of top 10%	-0.24% / -0.54%	-0.8 / -1.2	same picture
Price at least \$10 / volume at least \$5M	-0.22% / -0.21%	-0.7 / -0.8	same picture
Include "unclear" filing times	-0.38%	-0.8	same picture
Placebo: same stocks, 60 days earlier, no news	-0.40%	-0.4 (-1.2 to +0.3)	B trails A even without news: the gap is mechanical
Mirror on the short side (worst 10%, break below the low)	-0.50%	-1.1	symmetric; borrowing costs ignored
Friday announcements (22% of cases)	-0.16%	-2.8 (-4.7 to -0.8)	wrong sign for an "inattention" story; one split of many
Least / middle / most liquid third	-0.48% / -0.05% / -0.49%		not an illiquid-stock effect
Close near day-0 low / middle / high	-0.70% / -0.31% / -0.01%		B loses most where the high really binds

Costs. Estimated bid-ask spread (from daily highs and lows): median 0.82%, mean 0.96%. Half the spread plus 5 bp per side is about 1.06% per round trip.

	Before costs	After costs
Rule A, 2005 to 2026	+0.55%	-0.52% (-1.55 to +0.52)
Rule A, 2005 to 2014	+1.43%	+0.43% (-0.43 to +1.35)
Rule A, 2015 to 2026	+0.12%	-0.97% (-2.37 to +0.45)
Rule B, 2005 to 2026	+0.21%	-0.45% (-1.01 to +0.14)

Nothing here is tradable.

(Full table: *Appendix Table A1, outputs/tables/robustness_W10_H40.csv.*)

(Full table: *Appendix Table A2, outputs/tables/costs_W10_H40.csv.*)

10. What changed from the proposal, and why

#	Proposal said	Found	Tested by then	Did instead	Why better	Lesson
1	time each report with Massive 8-K timestamps	no time field; tags start 2020 and miss most releases	nothing	SEC EDGAR acceptance times	exact, public, point-in-time	read the field list, not the endpoint name
2	top 10% within each quarter	uses the future	nothing	top 10% of the previous 4 quarters	knowable on the day	selection must be computable at trade time
3	B buys "once it closes above the high"	cannot buy at a close you have not seen	nothing	buy next open; signal-close version as a check	every fill uses a real later price	fix fills before hypotheses
4	B beats A shows late drift	A contains B plus the run-up (section	nothing at design; one run of the first	per-day drift after breakout vs same	compares like with like	write the algebra before the code

#	Proposal said	Found	Tested by then	Did instead	Why better	Lesson
		2)	replacement	days, all cases		
5	Ticker Events for delistings	only symbol changes	probe	dead tickers from All Tickers, matched by CIK	keeps 22% of the sample	survivorship protection must be built
6	tick-level quotes for spreads	too much data, not needed	probe	Corwin-Schultz spread from daily high/low	computable for every event	use what is downloaded
7	Benzinga / income statements as second surprise	Benzinga not entitled (403)	probe	price reaction only, two alternative cut-offs	it is the variable the rule is about	probe entitlements first
8	adjust for dividends too	a few bp over 40 days; cancels in same-day comparison	nothing	splits only; noted as a limit	adjusts for what matters	match adjustment to horizon

Change 4 has one more step. The plan first replaced "B beats A" with "per-day return after the breakout vs before it". One run (02:13, experiment E00) showed the "before" part is meaningless: before a breakout the stock is, by definition, going up (+2.28% a day). So, before running the grid or any check, I switched to the same-days comparison above. Both versions are in the logs and tables; nothing is hidden.

11. Limits

- About one event in five is dated a day late because the 8-K lags the press release. This adds noise to A and B alike.
- Exchange membership is today's, not as of the event date.
- Returns are price returns above SPY; no dividends, no beta adjustment. The main test compares stocks with themselves on the same days, so this matters little.
- Costs are estimates, not fills.
- Confidence ranges rest on 86 quarters, not 14,770 events, so they are wide. The main test could detect an extra drift of about 0.7 bp a day (about 0.25% over the window); smaller effects cannot be ruled out.
- Ten design choices were made; only W and H were tuned, on 2005 to 2014, and reported on 2015 to 2026.

12. Conclusion

1. **No extra drift after a breakout:** 0.9 bp a day versus 1.8 for the same stocks on the same days; difference -0.9 bp (-1.5 to -0.2).
2. **Waiting costs money:** B trailed A by 0.34% overall and 0.92% before 2015.
3. **The earnings-day high is not special:** other price levels, a fixed delay and a no-news placebo all behave like B.

4. **The drift itself is small, old and untradable:** +1.4% before 2015, nothing since, and about 1% per round trip in costs.

In the brief's words: for the timing claim, *the original result was caused by a methodological issue*; for the drift, *it exists before costs but is not practically tradable*, and only before 2015.

13. What I would do next

- Date events by press-release time and measure how much the 8-K lag matters.
- Add a fundamental surprise (reported vs expected earnings) next to the price reaction.
- Test anchoring where the literature puts it: nearness to the 52-week high before the report.
- Compare with non-earnings days that had equally big moves, to separate earnings drift from ordinary momentum.

Glossary

Term	Meaning
Abnormal return	the stock's return minus SPY's return over the same days
Day 0	first trading session that could react to the report
Breakout	first close above the day-0 high, within W days
bp (basis point)	0.01%. 100 bp = 1%
Look-ahead	using information that was not available on the day
Survivorship bias	studying only companies that still exist
95% range (confidence interval)	values consistent with the data; includes zero = no clear effect
Season resampling	rebuilding the sample by drawing whole quarters at random, many times, to get the range
Calendar-time book	each day, the average return of all positions open that day
Newey-West t	a t-statistic corrected for day-to-day correlation
Corwin-Schultz spread	bid-ask spread estimated from daily highs and lows
Point-in-time	using, for each date, only what was known on that date

Questions an interviewer may ask

1. **Why did B lose to A before 2015?** On breakout cases A holds B's position plus a +3.3% run-up; the 40% that never broke out lost only 2.7% under A. B's zeros do not make up the gap.
2. **Are day-1 breakouts trivial?** Yes, but excluding them changes nothing (-0.37% vs -0.34%), and B loses most where the close was far below the high.
3. **Is the main test biased by picking breakouts?** No. It starts after the breakout day and compares with all strong-report stocks over the same days in the same quarter. The spike day is excluded on both sides.
4. **What would have convinced you?** A positive main test with a range above zero in both halves, controls that did not copy it, and a flat placebo. None happened.

5. **Where was the look-ahead in the proposal?** Ranking within the quarter, and buying at the signal close. Both were replaced before any test.
6. **Could late dating explain the null?** It adds noise, not bias; 64% of events are dated right and most of the rest are one day late.
7. **Why SPY as benchmark?** Simplicity. The main test compares stocks with themselves on the same days, so the benchmark cancels; the placebo is a second benchmark-free check.
8. **Is anything tradable?** No. Costs of about 1% per round trip exceed the +0.55% return; before 2015 the net is +0.43% with a range that includes zero.

References

Ball & Brown (1968) *JAR* 6, 159-178. Bernard & Thomas (1989) *JAR* 27 Suppl., 1-36; (1990) *JAE* 13, 305-340. Chan, Jegadeesh & Lakonishok (1996) *JF* 51, 1681-1713. Corwin & Schultz (2012) *JF* 67, 719-760. DellaVigna & Pollet (2009) *JF* 64, 709-749. Foster, Olsen & Shevlin (1984) *Accounting Review* 59, 574-603. Frazzini (2006) *JF* 61, 2017-2046. George & Hwang (2004) *JF* 59, 2145-2176. Grinblatt & Han (2005) *JFE* 78, 311-339. Hirshleifer, Lim & Teoh (2009) *JF* 64, 2289-2325. Huddart, Lang & Yetman (2009) *Management Science* 55, 16-31. Jegadeesh (1990) *JF* 45, 881-898. Lehmann (1990) *QJE* 105, 1-28. Martineau (2022) *Critical Finance Review* 11, 613-646.

Reproducing

`scripts/run_all.py` runs every stage. `scripts/build_notebook.py` rebuilds this document (notebook and markdown) from the committed tables alone. `python -m pytest -q` runs the checks.

Appendix: full tables

Every table below is a committed CSV in `outputs/tables/` and is also shown, with the code that loads it, in `notebooks/report.ipynb`.

Table A1. All checks (2005 to 2026)

Source: `outputs/tables/robustness_W10_H40.csv`

variant	cases	B minus A	lo	hi	extra drift/day	lo.	hi.
base top-decile trailing threshold	14769	-0.00338752	-0.00872638	0.00201595	-8.54602e-05	-0.000151757	-1.95192e-05
grid-selected cell W=20 H=60	14769	-0.00106196	-0.00643875	0.00442777	-3.68348e-05	-8.32195e-05	7.45476e-06
trigger from day 2 (exclude day-1 breakouts)	14769	-0.00370174	-0.00936693	0.00205901	-9.44503e-05	-0.000162851	-2.41837e-05

variant	cases	B minus A	lo	hi	extra drift/day	lo.	hi.
market-on-close entry for signal rules	14769	-0.00407551	-0.00934094	0.00133447	-0.00011566	-0.000182208	-5.02454e-05
selection: fixed ARO >= 5%	26610	-0.00183616	-0.00653123	0.00290047	-7.18412e-05	-0.000119879	-2.3031e-05
selection: z0 >= 2	35515	-0.000669464	-0.00442222	0.00343747	-4.67411e-05	-7.85392e-05	-1.27018e-05
intraday filings included	16025	-0.00381332	-0.0087933	0.00123643	-8.10377e-05	-0.000141404	-2.13777e-05
placebo: pseudo day 0 = 60 sessions before the event	14744	-0.00402525	-0.00905147	0.000984825	-4.34127e-05	-0.000115132	3.00655e-05
short-side mirror (bottom decile, close < day-0 low), gross	14767	-0.00500591	-0.00979412	5.96633e-05	-0.000108976	-0.000185794	-3.31379e-05
selection: trailing 85th percentile	21978	-0.002394	-0.0072423	0.00257764	-7.86907e-05	-0.000130349	-2.55322e-05
selection: trailing 95th percentile	7561	-0.00539022	-0.0110199	0.000391727	-0.000121054	-0.000207049	-3.86035e-05
filter: price >= \$10 on day -1	12624	-0.0022306	-0.00669892	0.00245388	-7.0377e-05	-0.000127282	-1.37012e-05
filter: median dollar volume >= \$5M	10816	-0.00209579	-0.00731453	0.00332758	-7.69025e-05	-0.000148625	-7.15867e-06
Friday day 0	3213	-0.00162313	-0.00767055	0.00420494	-0.000281813	-0.000473176	-7.73107e-05
Mon-Thu day 0	11556	-0.00387809	-0.00942435	0.00176763	-3.4018e-05	-0.000111942	4.05567e-05
liquidity tercile low	4924	-0.00477998	-0.0119341	0.00220579	-3.45304e-05	-0.000189127	0.000127384
liquidity tercile mid	4923	-0.000508767	-0.00708132	0.00603329	-0.000100306	-0.000261867	5.95112e-05
liquidity tercile high	4922	-0.00487383	-0.00964334	0.000270814	-0.000118715	-0.000280031	5.15665e-05
range-placement tercile low	4924	-0.00695953	-0.0148437	0.000909633	-0.000182511	-0.000354187	-1.23467e-05
range-placement tercile mid	4923	-0.00314292	-0.00854236	0.00234558	2.59475e-05	-0.000118675	0.000176697
range-placement tercile high	4922	-5.87124e-05	-0.00456996	0.00458003	-0.000121153	-0.000258413	6.61162e-06

Table A2. Costs

Source: outputs/tables/costs_W10_H40.csv

group	n	median_spread	gross_A	net_A	net_A_lo	net_A_hi	gross_B	net_B
all	14769	0.82%	+0.55%	-0.52%	-1.55%	+0.52%	+0.21%	-0.45%
selection_2005_2014	4803	0.75%	+1.43%	+0.43%	-0.43%	+1.35%	+0.51%	-0.11%
holdout_2015_2026	9966	0.86%	+0.12%	-0.97%	-2.37%	+0.45%	+0.06%	-0.62%
liq high	4922	0.73%	+0.59%	-0.39%	-1.52%	+0.69%	+0.10%	-0.54%
liq low	4924	0.92%	+0.86%	-0.29%	-1.44%	+0.92%	+0.38%	-0.31%
liq mid	4923	0.81%	+0.19%	-0.86%	-2.02%	+0.34%	+0.14%	-0.49%